

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
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Section / Batch:	BTECH IT	Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Distributed Database Query Processor

A distributed database processes queries using the recurrence

$T(n)=5T(n/2)+n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection.

Analyze how this impacts performance for large distributed systems.

Interpret efficiency using asymptotic notation.

Q1. Distributed Database Query Processor

Given the recurrence relation:

$$\left[\begin{array}{l} T(n) = 5T\left(\frac{n}{2}\right) + n \end{array} \right]$$

We solve it using the **Master Theorem**.

Step 1: Identify the Parameters

The standard form of the Master Theorem is:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Compare with the given recurrence:

$$T(n) = 5T\left(\frac{n}{2}\right) + n$$

Therefore,

- **a = 5**
- **b = 2**
- **f(n) = n**

Step 2: Calculate $(n^{\log_b a})$

$$n^{\log_b a} = n^{\log_2 5}$$

Since

$$\log_2 5 \approx 2.32$$

Therefore,

$$n^{\log_2 5} = n^{2.32}$$

Step 3: Compare $(f(n))$ with $(n^{\log_b a})$

Here,

$$f(n) = n = n^1$$

Since

$$[n = n^1]$$

and

$$[1 < 2.32]$$

we have

$$[f(n) = O(n^{2.32 - \epsilon})]$$

for some ($\epsilon > 0$).

Hence,

$$[f(n)]$$

is **polynomially smaller** than

$$[n^{\log_2 5}].$$

Step 4: Select the Master Theorem Case

This satisfies **Case 1** of the Master Theorem.

Case 1 Statement:

If

$$[f(n) = O(n^{\log_b a - \epsilon})]$$

for some ($\epsilon > 0$),

then

$$[T(n) = \Theta(n^{\log_b a})]$$

Step 5: Final Time Complexity

Therefore,

$$[\boxed{T(n) = \Theta(n^{2.32})}]$$

Impact on Large Distributed System

- The query is divided into **5 subqueries** at each recursion level.
- Each subquery processes **half the data**.
- The work grows faster than quadratic because the branching factor (**5**) is high.
- As the database size increases, query execution time increases rapidly.
- For very large distributed databases, this algorithm is **not highly scalable** because the computational cost grows significantly.

Efficiency Using Asymptotic Notation

- **Big-O:** ($O(n^{2.32})$) → Upper bound
- **Big-Theta:** ($\Theta(n^{2.32})$) → Tight (exact) bound
- **Big-Omega:** ($\Omega(n^{2.32})$) → Lower bound

Hence, the algorithm has **polynomial time complexity** of order **2.32**, making it less efficient than linear or ($n \log n$) algorithms for very large distributed systems.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____